

Experimental Verification of Automatic Sorting Machine Using Infrared Proximity Sensor and Microcontroller

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Abstract—This article proposes the design and builds of an automatic marigold (economic crop) sorting machine by sorting sizing 5 sizes with working together between conveyor belts, electric motors, and microcontroller control unit. The system designed to connect to WIFI to store data on Cloud computing and can display the information through the screen in Real-Time in numeric data format and graphs, Test the accuracy of the system by comparing experiments between microcontroller control unit and PLC control unit, which sold in the market at a high prices but this designed system has a low price and equivalent performance. The results from repeated tests showed that the automatic marigold sorting machine can operate as specified according to the design by separating the size of the marigolds.

Keywords—automatic, sorting machine, proximity sensor, microcontroller

I. INTRODUCTION

Marigold is an economic plant. There are high market demand and price is stable, can be sold throughout the year and it is a short-lived plant, takes 60 days can picking marigold flowers for sale and can be perennial, flowering for approximately 1 month. Marigold is an economic plant, therefore can create a career for people in the community[1]. For an area of 1 Rai, 5,000 marigolds can be planted. Takes time to care for marigolds plant since the beginning of planting takes 60 days and then from day 61- 120 can be cut marigold flowers other 15-times and 30,000 marigolds can be harvested in each time. There are sizing less than 4 cm, 4-5 cm, 6-7 cm, 8- 9 cm, and 10-12 cm, which the varying prices as shown in Table I. Causing marigold growers to have an income every 3-4 days, While 1 Rai can harvest the produce 15 times. The total duration of planting 1 Rai of marigold flowers is 120 days and the farmers will earn 30,000 Baht/Rai. In 1 year can harvest 3 times, therefore the area of 1 Rai able to generate income up to 120,000 Baht/Rai [2]

TABLE I. TABLE TYPE STYLES

Number	Size (cm)	Price (Bath)
1	10-12	1.00
2	8-9	0.50
3	6-7	0.25
4	4-5	0.10
5	4>	0



From the study, it was found that the number of marigolds collected by farmers each day was high. Size screening is necessary before selling it to the market. For this reason, the researcher has the idea of studying an automatic marigold sorting machine by relying on the motor to drive a conveyor when the object moves to the object detection point and the sensor unit will send signals to the PLC [3,4] control device to processing and after that, the PLC will be the transmitter ordered the air cylinder to work by pushing the object down the specified slot, but the PLC control unit has a quite high cost, therefore, there is a low-cost control unit design that can be used as practically equivalent to the PLC and farmers can use it on their farms at an affordable price, there is also a storage data system for verification so it is suitable and convenient for users by bringing the LOT system and Cloud Computing System for monitoring and processing the number of workpieces on a computer, mobile and tablet quickly and can also store data throughout the year for the farmers enable to analyze the data to compare the products in each season as well and also able to determine the stock of the product.

II. THEORY AND WORK PRINCIPLES

A. Propulsion system

For propulsion, by the conveyor and pulley to transport the objects from the source to the destination, the conveyor belt is commonly used such as flat belt, conveyor-belt and the v- belt conveyor, etc. This experiment uses 2 belt types which are:-

- Conveyor-belt, the material is a PVC green color, waterproof, thickness 3 mm, a width 200 mm, a total length of 5.8 m. used for supporting objects to be size sorting, a medium level of durability, able to support the moderate weight.
- V-Belt conveyor [5,6] the belt helps to transfer power from pulleys to the roller to drive the conveyor using the V-Belt, A-type.

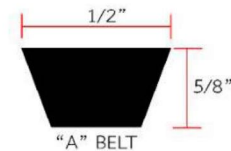


Fig. 1. V-Belt conveyor



Fig. 2. Size of belt A-type

Pulley is an important part of the belt conveyor system that acts in changing directions and control the tension or slack of the conveyor system. As for the motor used as an AC Motor, the rotation of the motor causes the belt to move forward to transport the object from the source to the destination.

The study of devices that will help to control the speed of the motor, Therefore, AC Motor Speed Controller is the equipment chosen for the increase and decrease the voltage to get the required speed in which the resistors can be adjustable.



Fig. 3. Pulley and AC motor

B. Pneumatic

The pneumatic system is a power transmission system that uses air pressure to transmit power by devices such as cylinders that convert wind energy into mechanical energy. The advantage of the pneumatic system is the speed of work. Compressed air has a high operating speed, resistant to explosion because the wind is not combustible and does not explode, therefore it is safe. The equipment in the pneumatic system does not damage when used overpower. It can adjust the speed easily and conveniently to installation. The transfer of air can be easily transmitted through pipes or wind tubes over long distances. Used air can be released into the atmosphere immediately without causing pollution. The air cylinders used in this experiment are two-way cylinders by air using a solenoid valve size 5/2 to control the wind direction.

C. Detection systems or sensors

For the selection of sensors, many factors are to be considered such as the purpose of the work. The worthiness of sensor, controller and sensor compatibility [7] therefore, in this work, choose to use Infrared Proximity Sensor. This sensor is a Diffuse Mode or directly reflects the object by using the principle of refraction of light and reflection with Emitter or Led Infrared as the signal to the receiver or the phototransistor is the receiver to the PLL circuit or Phase Lock Loop. Is an interference filter circuit or disturbance signal and sends it to the output. The object detection only occurs if the object moves nearby the sensor unit which has the spread of the electromagnetic field which generated by the RC oscillator circuit and when the capacitance changes to the one value of capacity will cause a condition called RC resonance which results in the occurrence oscillation of the signal as shown in Fig. 4.

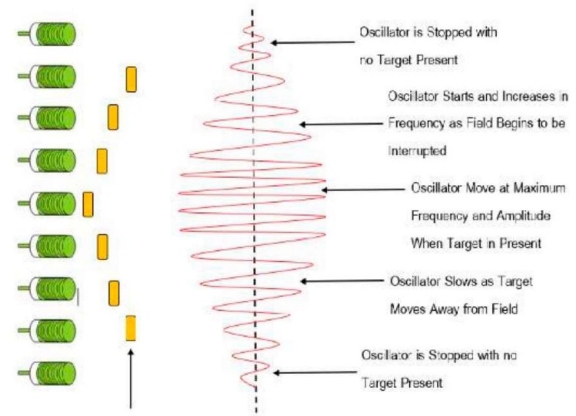


Fig. 4. Range of oscillators while detecting objects of the Proximity sensor

III. THE EXPERIMENTAL

Fig. 5. shows an overview of a control system created to sort the size of marigolds. The conveyor will be driven by AC Motor and take the object along the belt to the detection point where the Infrared Proximity Sensor has installed with a total of 4 stations, each station has been set the height of the detection level at 4 levels from the object release point is 10 cm, 8 cm, 6 cm, and 4 cm. for the last level no sensor installed because the object that can reach to the end is a non-standard sizes that is less than 4 cm in size. For the electricity source that is converted from 220 Volt to direct current 24 Volt to supply to the solenoid valve. For the motor, it has installed devices that will help in controlling the motor speed, it uses an adjustable resistor to increase and decrease the voltage before sending it to AC Motor.

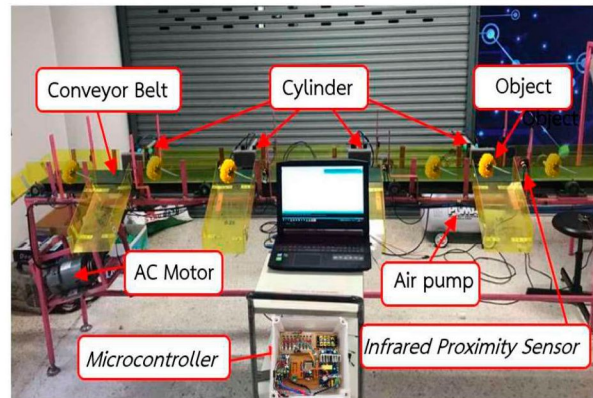


Fig. 5. Automatic size sorter

In the control box section, there are 4 main parts. The first part is the ESP32 board which is used to receive the signal from the sensor and forward it. The second part is the OPTO board which is the signal amplifier used to receive the signal from ESP32 sent to the valve to continue working. The third part is the part of the motor speed controller for adjusting the speed of the belt. The last part is the power supply used to supply power to support the system to operate the control box as shown in Fig. 6.

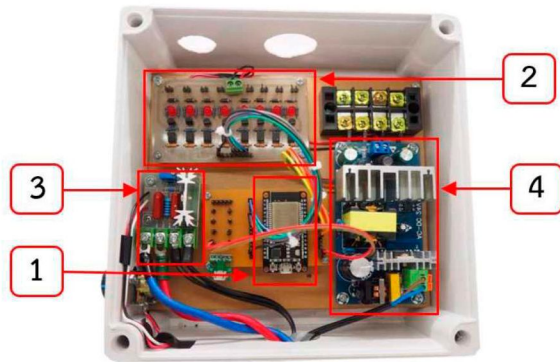


Fig. 6. Microcontroller control unit

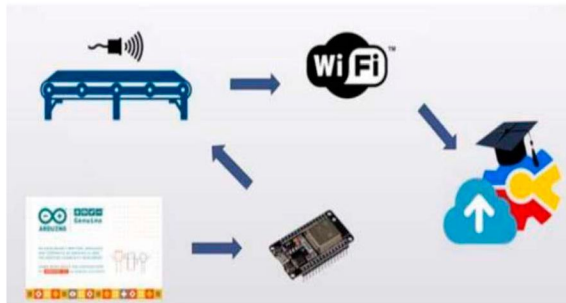


Fig. 7. Flowchart of the sorting machine

IV. STUDY RESULT

Results from the study showed that the size sorting 4 sizes of marigold and 25 flowers of each size by using an Infrared Proximity Sensor which uses multi-size detection of the ESP32 control unit can be used equivalent to a PLC control unit by showing the total workpiece table as shown in Table II and Table III.

TABLE II. RESULT OF CONTROL UNITS

Time (min)	PLC control			
	Size (cm)			
	Size A 10-12	Size B 8-9	Size C 6-7	Size D 4-5
5	8	3	7	4
10	13	10	16	12
15	19	17	21	17
20	25	25	25	25

TABLE III. RESULT OF THE EPS32 CONTROL UNITS

Time (min)	EPS32 control			
	Size (cm)			
	Size A 10-12	Size B 8-9	Size C 6-7	Size D 4-5
5	6	8	7	8
10	14	13	15	14
15	20	19	21	19
20	25	25	25	25

From the table shows that in each period of 5-20 minutes, different sizes of marigolds were detected between the PLC control unit and the EPS32 control unit. Due to the size of the marigolds tested were mixed in size without screening which

is not the same as the testing of the 2 control sets. However, the test results at the last minute, the automatic.



Fig. 8. Graph of experimental results of PLC control units



Fig. 9. Graph of experimental results of the ESP32 control unit



Fig. 10. The picture shows the sorting of marigold flowers.

Comparison of experimental result graphs between the PLC control unit and ESP32 control unit shown in each period of the size sorting of marigolds in 4 sizes and 25

flowers of each size. At the last moment, the marigold sizes can be sorted with the same results. Therefore, it can be demonstrated that the EPS32 control unit can function as a PLC control unit at a lower cost[8,9,10].

V. SUMMARY OF RESULTS

- Objects to be tested able to sort the size successfully, Sensors for sorting, conveyor belt, and other devices can work correctly according to the model placed.
- From the test of the size sorting of objects, it is found that the ESP32 control unit can be used equivalent to the PLC control unit.
- The part of programming (Software) has been experimented with separating the size of the object by the real machine and found that the size of the object can be separated as needed, also can connect to WIFI to store data to the Cloud and can display the information on the screen in Real-Time both in numeric data format and graphs as well.

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